

BuildMarshalAI

Construction document intelligence that **refuses to guess**. Every answer points at the page it came from; every field it cannot read, it asks for.

Md. Mobasharul Islam (2105076) · [Team member 2] · [Team member 3] | Supervised by [Supervisor name], Department of Computer Science & Engineering



ColPali v1.2 · MaxSim
 ChromaDB · FastAPI
 Gemini via CLIPProxyAPI
 1 × RTX 3070 · on-premises

A construction contract is a legal document. Slabs are specified at 3500 psi and piles at 4000 psi — an assistant that confuses the two has not made a typo, it has **created a liability**.

Site engineers still answer such questions by hand-searching hundreds of pages of specifications and drawings. A general-purpose assistant answers **fluently and wrongly**, with nothing to check it against — and will invent a project, a date or a person to finish filling a form.

GUARANTEE 01 — READING

Every answer points at the page it came from.

Retrieval runs over **rendered page images**, not extracted text. A ColPali v1.2 vision-language encoder embeds each page as a grid of patch vectors and MaxSim scores the question against them — so a table, a stamped revision or a floor plan is searchable on the same footing as a paragraph.

- The citation is part of the answer.** Document, page and MaxSim score travel with every reply; one click opens that exact page image. Scanned drawings and dense tender tables — where a text-chunk pipeline returns nothing usable — are retrieved the same way.
- Nothing found means nothing said.** With no passage above threshold it says so and names what the workspace holds, rather than composing something plausible.

The answer alongside is checkable in one click: **06_Tender-Specification.pdf, page 2, 94% MaxSim** — the page that actually carries the concrete schedule.

* Grade beams — 4000 psi, 50 mm cover
* Slabs and beams, Levels 1-5 — **3500 psi (C25/30)**, 25 mm cover
* Columns — 4000 psi, 40 mm cover

All concrete must come from a BSTI-accredited ready-mix plant, with one set of cylinder tests per 50 cubic metres or per pour, whichever is more frequent.

1

06_Tender-Specific...
Page 2
94%
Evidence

01:40 AM

A live answer and its **evidence card** — grades differ by location.

GUARANTEE 02 — WRITING

Every field it cannot read, it asks for.

Chat creates **eleven kinds of record** — projects, tasks, users, roles, costs, procurement, catalogues. What each one mandates is derived from the application's own constructors and validators, so the assistant and the API can never disagree about what is required.

- A reference must exist.** A project type, trade or assignee is accepted only when it names a record already in the workspace; an unmatched value is dropped and asked about, never corrected to the nearest guess.
- It collects across turns.** Values given one message at a time accumulate; it asks only for what is outstanding and never re-asks.
- The form opens once, at the end.** Complete and prefilled for review — then the record is written by the `POST` route that already owns the permission check and the validator.

Interpretation creates nothing. **529 tests / 591 cases** hold the line — including that an unreadable date is asked for again rather than filled in.

Got it. I still need **Manager** and **Start date**.

So far: **Project code:** WR-2027 · **Project name:** Website Redesign

01:41 AM

John is the project manager

01:41 AM

Got it. I still need **Start date**.

So far: **Manager:** John · **Project code:** WR-2027 · **Project name:** Website Redesign

01:41 AM

Mid-collection: each reply names **only what is still missing**.

Project Name
Website Redesign

Project Code
WR-2027

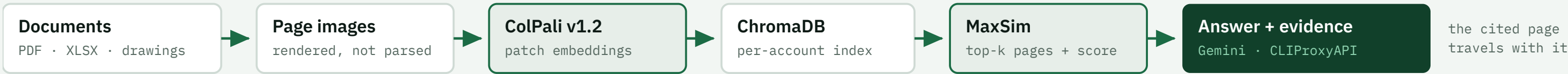
Project Manager
John

Type
— Select —

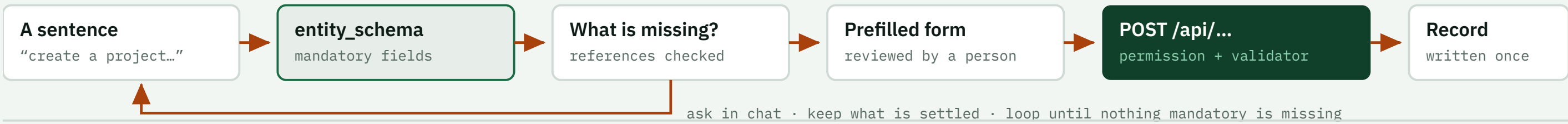
Then the dialog opens **once, prefilled**, for review.

HOW IT WORKS — TWO PATHS, ONE GUARANTEE EACH

READING · A QUESTION



WRITING · A COMMAND



Each account holds its own documents, index and settings — isolation is structural, not a filter. · Retrieval runs locally; pages are never uploaded to search them.

THE REST OF THE PLATFORM

- Document onboarding**
A programme or cost plan becomes a reviewable draft — you confirm before anything is committed.
- Google & Microsoft 365**
Drive indexing, mail drafting, calendar and Meet — every booking behind an explicit confirm.
- Structural multi-tenancy**
Each account keeps its own documents, index, projects and settings — nothing is shared by default.
- Earned value & flow**
SPI, CPI and the 50/50 rule computed from the live programme, with rule-based risk flags.
- Generated reports**
Phase-by-phase project reports rendered to PDF from the same cited evidence.
- Evidence viewer**
Open the cited page itself, at the region that matched, without leaving the answer.

BUILT AND TESTED

591 test cases passing, from 529 test functions	112 API routes across 23 backend modules	11 record types creatable from a sentence
26k lines of Python and JavaScript	0 invented references accepted, by design	1 consumer GPU — runs fully on-premises

